Day 02 | Friday

October 30, 2026

Quantities, averages and word problems

9:00 AM–10:15 AM • Math • 75 min

Cut-list practice and rounding

10:30 AM–12:00 PM • Calculation Practice • 90 min

Week 1 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Quantities, averages and word problems

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:25 AM	**Teach**	15
9:25 AM–9:40 AM	**Together**	15
9:40 AM–10:00 AM	**Practise**	20
10:00 AM–10:10 AM	**Check**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math • Math Week 1, Lesson 3

10:30 AM–12:00 PM • 90 min

Cut-list practice and rounding

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Calculation Practice • Math Week 1, Lesson 4

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Quantities, averages and word problems

Today’s goal: Choose operations from a job problem and explain the difference between full pieces and purchase quantities.

Name one use for today’s skill.

Textbook Chapter 1 pp.15–21

Exercise Book 1A Q15–19 (p.7); 1B Q2–5 (p.8)

Keep your notebook, assigned ACCESS books and chosen calculator ready.

Textbook Chapter 1 pp.15–21

Recall and reconnect

Identify what information is given, what must be found and which unit the answer needs.

Try first without displayed answers.

Resources: Exercise Book 1A Q15–19 (p.7); 1B Q2–5 (p.8)

Textbook Chapter 1 pp.15–21

Translate the story into groups

1. Write what is given and what must be found before choosing an operation.
2. Equal groups: number of groups × amount in each group gives the total.
3. Sharing equally: total ÷ number of groups gives the amount in each group.
4. Average: add all the values, then divide by how many values were added.

Textbook Chapter 1 pp.15–21

Separate need from purchase quantity

1. First calculate the exact number or length needed for the job.
2. Divide by the supplier’s quantity per box or stock length.
3. A part box still requires a whole box: round the purchase up.
4. Check how much the whole-unit order supplies and whether it covers the need.

Textbook Chapter 1 pp.15–21

Pieces needed ≠ boxes purchased

Classroom example: 12 brackets need 6 screws each. A box holds 20.

12 × 6 = 72 screws needed
72 ÷ 20 = 3.6 boxes → buy 4 whole boxes
Check: 4 × 20 = 80 screws, so 8 remain

Average example

Lengths 60, 64, 68 and 72 mm:
(60 + 64 + 68 + 72) ÷ 4 = 66 mm

Work together

Exercise Book 1B Q2–3. Separate the calculated need from the supplier order. An aggregate length estimate is not a verified cutting layout.

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Example 1.13 Average welds per day

Over five days a welder completes 38, 42, 45, 40 and 35 joints. What is the average number of joints per day?

SOLUTION

1. Add: 38 + 42 + 45 + 40 + 35 = 200.
2. There are 5 days, so divide by 5: 200 ÷ 5 = 40.

Textbook p.20 · Example 1.13: add the values, then divide by their count.

Textbook Chapter 1 pp.15–21

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Example 1.14 Ordering brackets

A job has 3 units. Each unit needs 14 brackets, and you want 6 spares. Brackets come in boxes of 10. How many boxes do you order?

SOLUTION

1. Brackets for the units: $3 \times 14 = 42$.
2. Add the spares: $42 + 6 = 48$ brackets.
3. Boxes: $48 \div 10 = 4.8$.
4. 4 boxes hold only 40 brackets, which is not enough. Round up to 5 boxes.

Textbook p.20 · Example 1.14: calculate need, then round the whole-box order up.

Textbook Chapter 1 pp.15–21

Spot the mistake. Explain the correction.

A job needs 3.6 boxes. Why is ordering 3 boxes not enough?

Discuss the method before checking the answer.

Three boxes provide less than the calculated need. Buy 4 whole boxes and check the quantity supplied.

Textbook Chapter 1 pp.15–21

Work together

Solve 1B questions 2–3. Label the number needed and the number of full supplier units separately.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 1A Q15–19 (p.7); 1B Q2–5 (p.8)

Textbook Chapter 1 pp.15–21

Guided practice | 1B Q2

A plumber measures five pipe runs: 12 ft, 9 ft, 15 ft, 7 ft, 18 ft. The supplier sells this pipe in 21 ft lengths. Ignore fittings and waste. How many feet of pipe are needed, and how many 21 ft lengths must be bought?

Work with a partner. Take turns explaining the next step.

Textbook Chapter 1 pp.15–21

Compare your method | 1B Q2

A plumber measures five pipe runs: 12 ft, 9 ft, 15 ft, 7 ft, 18 ft. The supplier sells this pipe in 21 ft lengths. Ignore fittings and waste. How many feet of pipe are needed, and how many 21 ft lengths must be bought?

CHECK TOGETHER

61 ft. 61 ÷ 21 = 2.9, so buy 3 lengths.

Round up: a part length still has to be bought.

Textbook Chapter 1 pp.15–21

Your scheduled practice

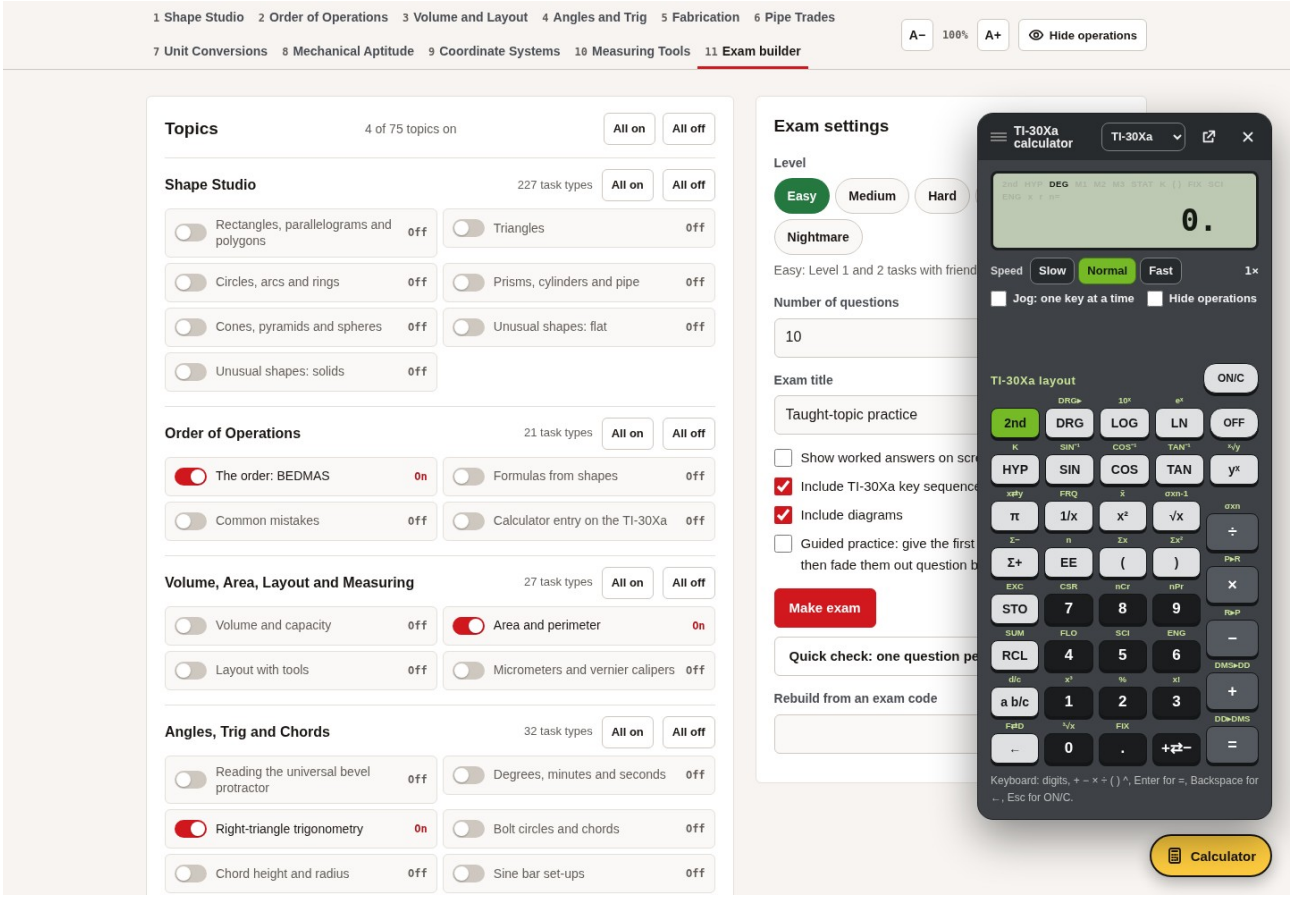
Complete 1A questions 15–18 and 1B questions 4–5. Explain each operation and check the answer using multiplication or addition.

Show working and units; estimate, then check.

Resources: Exercise Book 1A Q15–19 (p.7); 1B Q2–5 (p.8)

Textbook Chapter 1 pp.15–21

Open the ACCESS tool for this activity



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 1 pp.15–21

Your tool activity: try, compare and record

1. Supported: Draw the groups and label what each number counts in 1B Q2–3.

2. Independent practice: Write the operation and units for the assigned 1A/1B questions before keying.

3. Independent checking: Use Numeracy Hub Calculator, then verify with multiplication or addition.

Textbook Chapter 1 pp.15–21

Practise from the Exercise Book | 1A Q15

32×15

Write the operation, working and units before checking.

Textbook Chapter 1 pp.15–21

Practise from the Exercise Book | 1A Q16

$$215 \times 24$$

Write the operation, working and units before checking.

Textbook Chapter 1 pp.15–21

Check independently

Complete 1A question 19 and 1B question 3 independently.

Answer independently and show the method.

Resources: Exercise Book 1A Q15–19 (p.7); 1B Q2–5 (p.8)

Textbook Chapter 1 pp.15–21

Independent check | 1A Q19

Find the average of 64, 58, 71 and 67.

Answer independently. Keep worked solutions closed.

Textbook Chapter 1 pp.15–21

Independent check | 1B Q3

A millwright sets 14 baseplates. Each baseplate takes 8 anchor bolts. Anchor bolts come in boxes of 25. How many bolts are needed, and how many boxes?

Answer independently. Keep worked solutions closed.

Textbook Chapter 1 pp.15–21

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 1A Q15–19 (p.7); 1B Q2–5 (p.8)

Textbook Chapter 1 pp.15–21

Break

10:15 AM–10:30 AM

Return ready for: Cut-list practice and rounding

This break is excluded from teaching minutes.

Cut-list practice and rounding

Today’s goal: Solve multi-step stock problems and distinguish rounding up purchases from rounding down complete pieces.

Name one use for today’s skill.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Exercise Book 1B Q1–5 (p.8); 1C Q1–2 (p.10)

Keep your notebook, assigned ACCESS books and chosen calculator ready.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Recall and reconnect

Explain why buying boxes and cutting full pieces need different rounding decisions. Retrieve 1A Q18 (p.7).

Try first without displayed answers.

Resources: Exercise Book 1B Q1–5 (p.8); 1C Q1–2 (p.10)

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Recall from the Exercise Book | 1A Q18

Divide 5,000 by 320. Give the whole-number quotient and the remainder.

Write the operation, working and units before checking.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Account for the stock length

1. List the original stock and every specified piece in the same unit.
2. Add the piece lengths to obtain the total used.
3. Offcut = original stock − total used.
4. Check by adding used length and offcut to restore the original stock.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Let the question choose the rounding direction

1. Buying enough whole supplier units requires rounding up.
2. Counting complete pieces from fixed stock requires rounding down.
3. Calculate the offcut from the whole pieces actually obtained.
4. A total-length calculation alone does not prove a cutting arrangement will fit.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Account for the whole stock length

Exercise Book 1B Q1: stock 6,000 mm; pieces 1,350, 980 and 2,145 mm.

Used = 1,350 + 980 + 2,145 = 4,475 mm
Offcut = 6,000 − 4,475 = 1,525 mm
Check: 4,475 + 1,525 = 6,000 mm

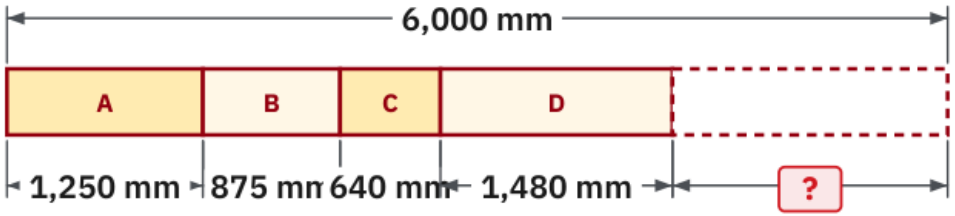
Complete-piece example, ignoring kerf: 100 in ÷ 18 in = 5 full pieces.
Offcut = 100 − 5 × 18 = 10 in. A sixth full piece does not fit.

Together: 1B Q2–3. Identify given quantities, choose operations and justify rounding. Add kerf only when stated; never assume a cutting layout is feasible from a total alone.

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Example 1.6 What is left of the bar?

The four pieces from Example 1.5 (4,245 mm in total) are cut from a 6,000 mm bar. Ignore the width of the saw cuts. How long is the offcut?



SOLUTION

1. Write 6,000 on top and 4,245 below it.
2. Ones: 0 − 5 needs a borrow. The tens and hundreds are 0, so borrow through them: 6,000 becomes 5 thousands, 9 hundreds, 9 tens and 10 ones.
3. 10 − 5 = 5, 9 − 4 = 5, 9 − 2 = 7, 5 − 4 = 1.

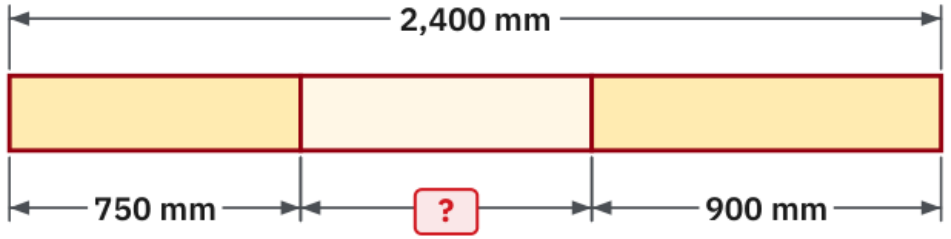
Textbook p.14 · Example 1.6: used length and offcut account for the original stock.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

ACCESS Textbook | page 14

Example 1.7 Finding a hidden dimension

A drawing gives the overall length of a part as 2,400 mm and two of its three sections as 750 mm and 900 mm. Find the middle section.



SOLUTION

1. The three sections add up to the overall length.
2. Add the sections you know: 750 + 900 = 1,650.
3. Subtract from the overall length: 2,400 − 1,650 = 750.

Textbook p.14 · Example 1.7: known parts and a hidden dimension.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Spot the mistake. Explain the correction.

100 in ÷ 18 in is about 5.56. Can the stock produce 6 complete 18 in pieces?

Discuss the method before checking the answer.

No. Six pieces need 108 in. Five complete pieces use 90 in, leaving 10 in, ignoring kerf.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Work together

Work Exercise Book 1B Q2–3 in pairs. Mark given quantities, choose operations, and justify purchase rounding.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 1B Q1–5 (p.8); 1C Q1–2 (p.10)

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Guided practice | 1B Q2

A plumber measures five pipe runs: 12 ft, 9 ft, 15 ft, 7 ft, 18 ft. The supplier sells this pipe in 21 ft lengths. Ignore fittings and waste. How many feet of pipe are needed, and how many 21 ft lengths must be bought?

Work with a partner. Take turns explaining the next step.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Compare your method | 1B Q2

A plumber measures five pipe runs: 12 ft, 9 ft, 15 ft, 7 ft, 18 ft. The supplier sells this pipe in 21 ft lengths. Ignore fittings and waste. How many feet of pipe are needed, and how many 21 ft lengths must be bought?

CHECK TOGETHER

61 ft. 61 ÷ 21 = 2.9, so buy 3 lengths.

Round up: a part length still has to be bought.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Your scheduled practice

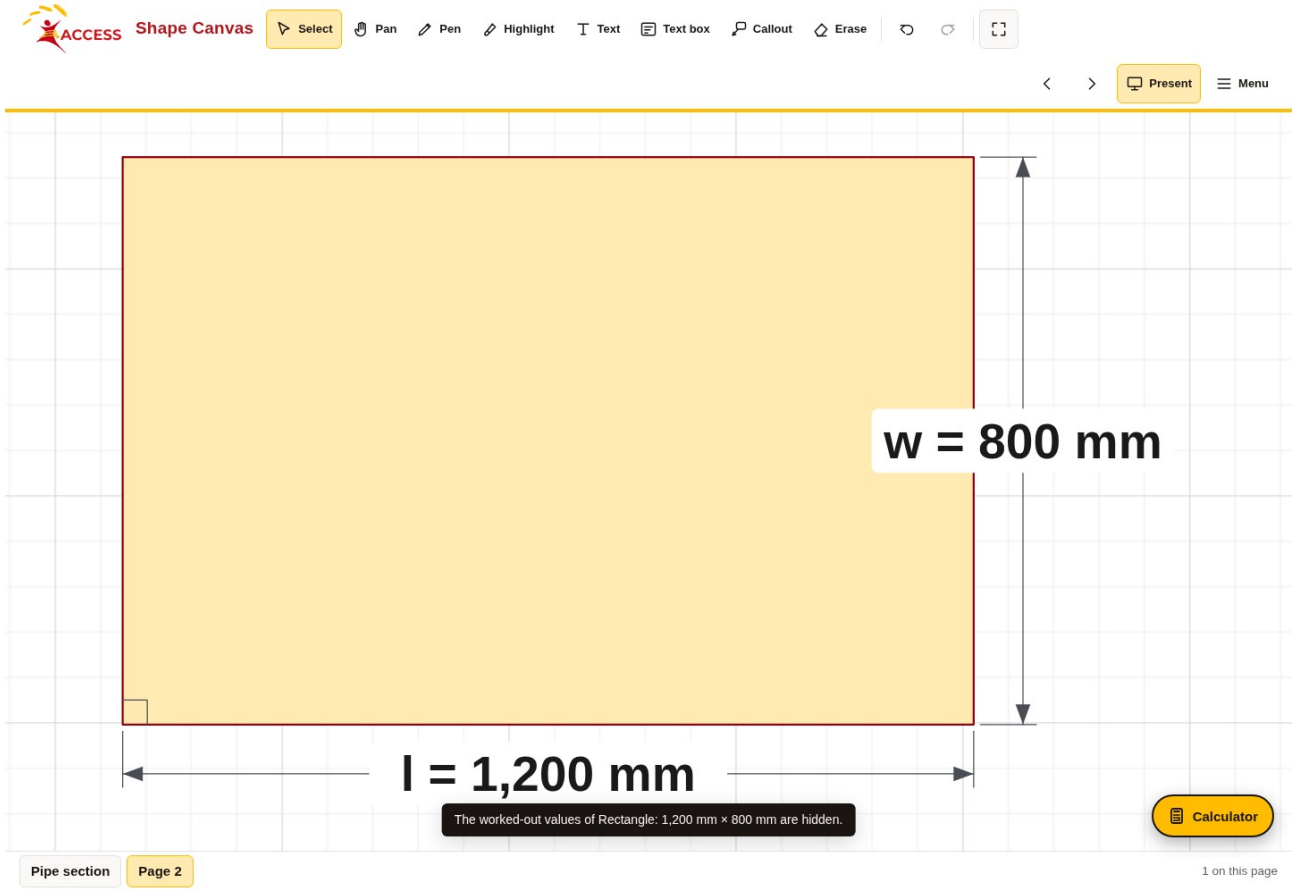
Solve Exercise Book 1B Q4–5 and 1C Q1–2. Recreate one cut-list sketch on Shape Canvas; write the equation before checking. Use extra time to correct and explain a second method.

Show working and units; estimate, then check.

Resources: Exercise Book 1B Q1–5 (p.8); 1C Q1–2 (p.10)

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Open the ACCESS tool for this activity



Navigation example. Use the values and units in the assigned task.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Your tool activity: try, compare and record

1. Supported: In Shape Canvas, add three rectangles for the cut pieces and label their lengths.

2. Independent practice: Add a Text label for the original stock; write the cut-list equation on paper.

3. Independent checking: Verify the totals in Numeracy Hub Calculator and reconstruct the stock length.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Practise from the Exercise Book | 1B Q4

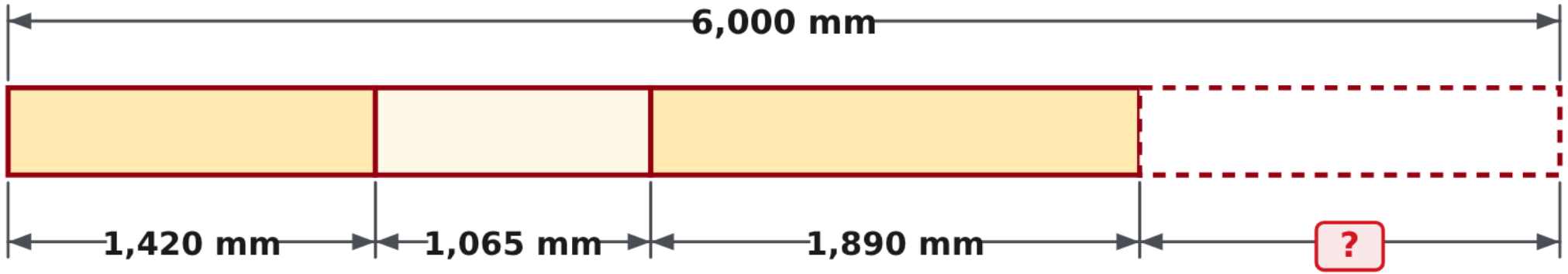
A pipe job has 9 flanged joints. Each joint takes 8 bolts and each bolt takes 2 nuts. How many nuts are needed?

Write the operation, working and units before checking.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Practise from the Exercise Book | 1C Q1

Three pieces are cut from a 6,000 mm bar. Ignore the saw cuts. Find the offcut.



Write the operation, working and units before checking.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Check independently

Rework Exercise Book 1B Q3 and Q5 independently; explain why the two rounding directions differ.

Answer independently and show the method.

Resources: Exercise Book 1B Q1–5 (p.8); 1C Q1–2 (p.10)

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Independent check | 1B Q3

A millwright sets 14 baseplates. Each baseplate takes 8 anchor bolts. Anchor bolts come in boxes of 25. How many bolts are needed, and how many boxes?

Answer independently. Keep worked solutions closed.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Independent check | 1B Q5

Lengths of 18 in are cut from a 20 ft bar of angle (20 ft = 240 in). Ignore the saw cuts. How many pieces can be cut, and how long is the offcut?

Answer independently. Keep worked solutions closed.

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 1B Q1–5 (p.8); 1C Q1–2 (p.10)

Textbook sections 1.4–1.8 pp.12–21; sections 2.11 p.39